Exercise 1: Parts Order Form

a. Assumption:
1. CustomerNumber uniquely identifies a customer, CustomerName and CustomerType depend on CustomerNumber.
2. A customer can place multiple orders over time, but each order is placed by one customer, and each customer places at most one order at a given date amd time.
3. An order is handled by one employee at a specific date and time.
4. An employee can handle multiple orders.
5. A part belongs to one type and is stored in one cage code.
6. A cage code represents a specific cage/shelf location for inventory.
7. PartNumber uniquely identifies a part ,PartName and PartType depend on PartNumber.
8. Unit price is determined by the part
9. Quantity ordered is specific to a particular line item in an order.

Exercise 1 -- First Normal Form (1NF)

customerNumber	**customerName**	**customerType**	**orderDate**	**orderTime**	**employeeName**	**partNumber**	**partName**	**partType**	**cageCode**	**quantityOrdered**	**unitPrice**
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10654	Float Control	Plumbing	G413	4	12
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

unnest the repeating parts group so that each row contains one part and all fields are atomic. This is 1NF, but it has partial dependencies because customer data depends only on CustomerNumber and part data depends only on PartNumber.

1NF primary key: **customerNumber, orderDate, orderTime, artNumber.**

Exercise 1 -- Second Normal Form (2NF)

customer

customerNumber	customerName	customerType
HG54587	Jeff Peterson	Consumer

remove attributes that depend on only part of the composite key.customerName and customerType depend only on customerNumber,

part

partNumber	**partName**	**partType**	**cageCode**	**unitPrice**
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

partName, partType, cageCode, and unitPrice depend on partNumber

purchase_order

customerNumber	**orderDate**	**orderTime**	employeeName
HG54587	7/1/2024	10:30 am	D. Harrison

Keep employeeName with the purchase_order

order_line

customerNumber	***customerNumber***	***orderTime***	***partNumber***	quantityOrdered
HG54587	7/1/2024	10:30 am	10654	4
HG54587	7/1/2024	10:30 am	10456	3
HG54587	7/1/2024	10:30 am	10776	7
HG54587	7/1/2024	10:30 am	10657	5

keep quantityOrdered in the order line because it depends on the full order-plus-part key.

Exercise 1 -- Third Normal Form (3NF)

customer

customerNumber	**customerName**	**customerType**
HG54587	**Jeff Peterson**	**Consumer**

order_line

customerNumber	***customerNumber***	***orderTime***	***partNumber***	quantityOrdered
HG54587	7/1/2024	10:30 am	10654	4
HG54587	7/1/2024	10:30 am	10456	3
HG54587	7/1/2024	10:30 am	10776	7
HG54587	7/1/2024	10:30 am	10657	5

part

partNumber	partName	*partType*	unirPrice
10654	Float Control	Plumbing	12
10456	Modulator	Electrical	7
10776	Hose Assembly	Plumbing	9
10657	Float Assembly	Plumbing	10

part_type

partType	cageCode
Plumbing	G413
Electrical	H433

extract cageCode from the previous part, and assign partType as its key

purchase_order

customerNumber	**orderDate**	**orderTime**	empolyeeName
HG54587	7/1/2024	10:30 am	D. Harrison

Foreign keys:
purchase_order.customerNumber → customer.customerNumber;
order_line.(customerNumber, orderDate, orderTime) →
purchase_order.(customerNumber, orderDate, orderTime);
order_line.partNumber → part.partNumber.

Exercise 2: Panacea Mental Health Corporation

a. Assumptions

1.staffNo uniquely identifies a therapist and determines therapistName.
2.patNo uniquely identifies a patient and determines patName.
3.a therapist cannot have two appointments at the same date and time, so (staffNo, appointmentDate, appointmentTime) identifies an appointment. A patient may have multiple appointments in a day, including with different therapists.
4.a therapist works at only one branch on a given date, so (staffNo, appointmentDate) determines branchNo
5. The branch is determined by the therapist's work location for that day
6. branchNo is treated as an identifier; no additional branch attributes are supplied.

Text

Exercise 2 -- First Normal form (1NF)

staffNo	therapistName	patNo	patName	**appointmentDate**	**appointmentTime**	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

The appointment date and time are separate atomic attributes. The given data can therefore be represented as one row per appointment.
Composite PK: (**staffNo, appointmentDate, appointmentTime**),

Exercise 2 -- Second Normal form (2NF)

therapist

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

therapistName depends only on staffNo, a proper subset of the appointment key.

therapist_day

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

branchNo depends on (staffNo, appointmentDate), also a proper subset of the appointment key.

appointment

staffNo	***appointmentDate***	**appointmentTime**	patNo	patName
S1011	9/12/2022	10:00	P100	Lily White
S1011	9/12/2022	12:00	P105	Jill Baker
S1024	9/12/2022	10:00	P108	Andy McKee
S1024	9/14/2022	14:00	P108	Andy McKee
S1032	9/14/2022	16:30	P105	Jill Baker
S1032	9/15/2022	18:00	P110	Jimmy Winter

keep patNo and patName in appointment temporarily; their dependency is transitive and will be addressed in 3NF.

2NF keys: therapist PK = staffNo. therapist_day PK = (staffNo, appointmentDate); staffNo is an FK to therapist.
appointment PK = (staffNo, appointmentDate, appointmentTime); (staffNo, appointmentDate) references therapist_day.
patNo is not part of the appointment PK under the stated no-overlapping-appointments assumption.

Exercise 2 -- Third Normal form (3NF)

therapist

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

therapist_day

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

patient

patNo	PatName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

patName depends on patNo rather than on the appointment key

appointment

staffNo	***appointmentDate***	**appointmentTime**	*patNo*
S1011	9/12/2022	10:00	P100
S1011	9/12/2022	12:00	P105
S1024	9/12/2022	10:00	P108
S1024	9/14/2022	14:00	P108
S1032	9/14/2022	16:30	P105
S1032	9/15/2022	18:00	P110

keep patNo as a foreign key in appointment

Foreign keys:
therapist_day.staffNo → therapist.staffNo; appointment.(staffNo, appointmentDate) → therapist_day.(staffNo, appointmentDate); appointment.patNo → patient.patNo.
The staffNo and appointmentDate fields in appointment form part of its PK and together are also an FK.

Exercise 3. Maid Better Temp Agency

a. Assumption:
1. eNo uniquely identifies an employee and determines eName.
2. contractNo uniquely identifies a contract and each contract applies to exactly one event, so contractNo determines eventNo.
3. eventNo uniquely identifies an event and determines eventLoc.
4. An employee may work on multiple contracts, and a contract may have multiple employees.The hours value records the hours worked by one employee on one contract, so (eNo, contractNo) is the composite primary key.
5. Contract number and event number are identifiers supplied in the data;

Exercise 3 -- First Normal form (1NF)

eNo	**contractNo**	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

Each row represents one employee's hours on one contract. All attributes are atomic.
PK: (eNo, contractNo).

Exercise 3 -- Second Normal form (2NF)

empolyee

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

eName depends only on eNo, and eventNo depends only on contractNo.
Both are partial dependencies on parts of the composite key (eNo, contractNo).
Move employee details into Employee and contract/event information into Contract.
Keep hours in the intersection relation because it depends on both eNo and contractNo.
eventLoc is temporarily retained in contract; its dependency on eventNo will be removed in 3NF.

eName depends only on eNo

contract

contractNo	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

eventNo depends only on contractNo.

employee_contract

eNo	***contractNo***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

2NF keys: employee PK = eNo. contract PK = contractNo.
employee_contractPK = (eNo, contractNo); eNo references employee and contractNo references contract.

Exercise 3 -- Third Normal form (3NF)

eventLoc depends on eventNo, not directly on contractNo.
Since contractNo → eventNo and eventNo → eventLoc, eventLoc is transitively dependent on the contract primary key.
Create event(eventNo, eventLoc), and keep eventNo as an FK in contract.

empolyee

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

event

eventNo	eventLoc
H25	Queens
H4	Yonkers

contract

contractNo	*eventNo*
C1024	H25
C1025	H4
C1026	H25

employee_contract

eNo	***contractNo***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

Foreign keys:
contract.eventNo → event.eventNo; employee_contract.eNo → employee.eNo;
employee_contractt.contractNo → contract.contractNo. In employee_contract, both foreign keys are also part of the composite primary key.